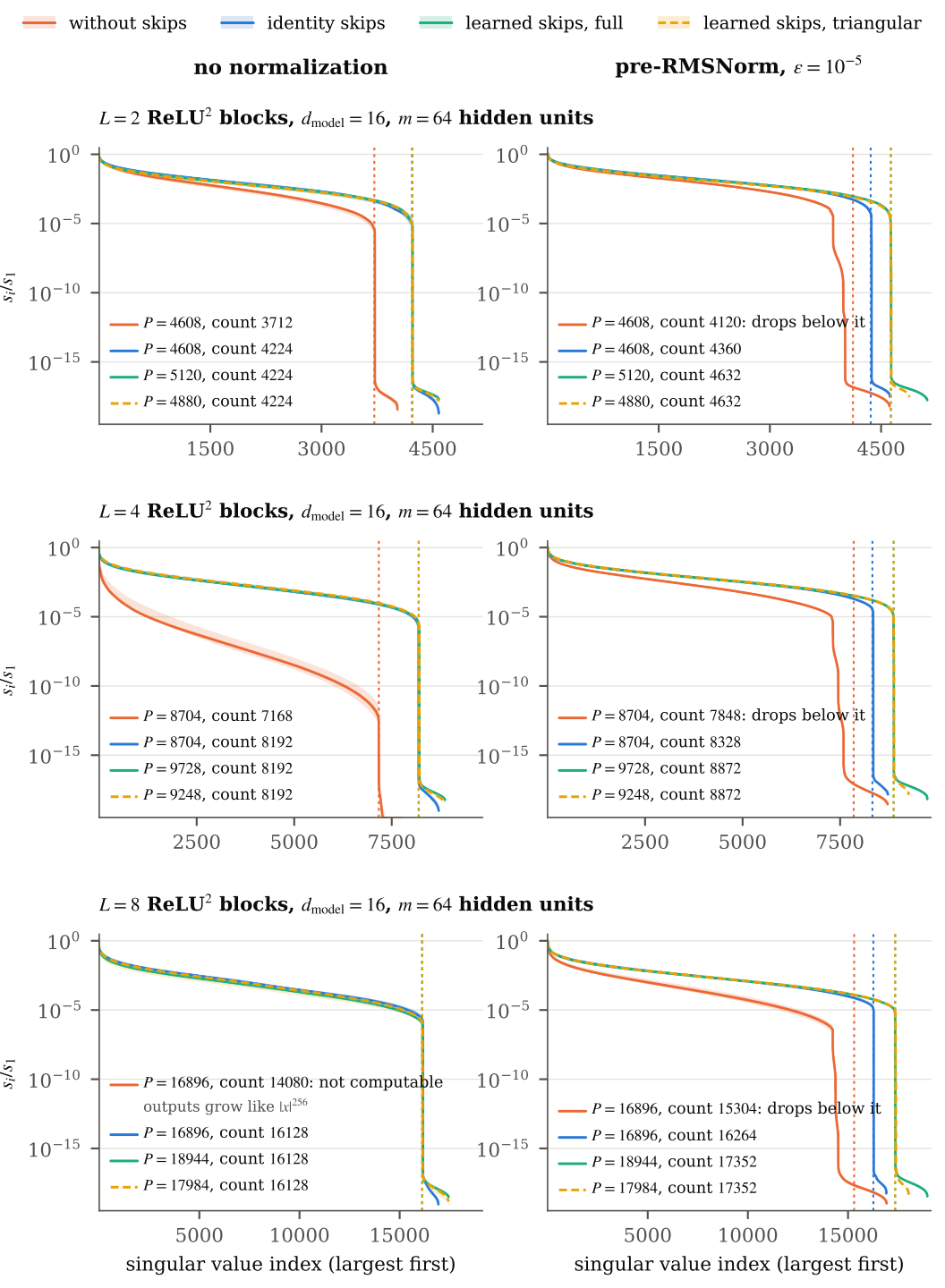

without skips
identity skips
learned skips, full
learned skips, triangular
no normalization
pre-RMSNorm, $\varepsilon = 10^{-5}$
$L = 2$ ReLU^2 blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
$L = 4$ ReLU^2 blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
$L = 8$ ReLU^2 blocks, $d_{\text{model}} = 16$, $m = 64$ hidden units
s_i/s_1
10^0
10^{-5}
10^{-10}
10^{-15}
$P = 4608$, count 3712
$P = 4608$, count 4224
$P = 5120$, count 4224
$P = 4880$, count 4224
$P = 4608$, count 4120: drops below it
$P = 4608$, count 4360
$P = 5120$, count 4632
$P = 4880$, count 4632
$P = 8704$, count 7168
$P = 8704$, count 8192
$P = 9728$, count 8192
$P = 9248$, count 8192
$P = 8704$, count 7848: drops below it
$P = 8704$, count 8328
$P = 9728$, count 8872
$P = 9248$, count 8872
$P = 16896$, count 14080: not computable
outputs grow like $|x|^{256}$
$P = 16896$, count 16128
$P = 18944$, count 16128
$P = 17984$, count 16128
$P = 16896$, count 15304: drops below it
$P = 16896$, count 16264
$P = 18944$, count 17352
$P = 17984$, count 17352
1500
3000
4500
2500
5000
7500
10000
15000
singular value index (largest first)